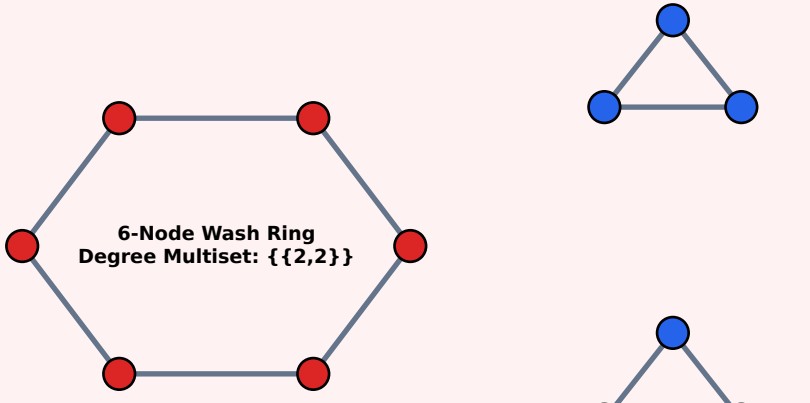
Standard Spatial GNN (1-WL Bottleneck: Identical Multisets)



1-WL COLOR REFINEMENT FAILS: c(u) = c(v)
Cannot distinguish circular wash cycle from benign bipartite transfers!

C-STGB Disambiguation: 12-D Universal Canonical Invariants

Kirchhoff Mass-Flow Invariant
Wash Ring: $\tilde{\Phi}_{\text{flow}} = 1.00 \pm 0.01$ (Conserved)
Benign Clusters: $\tilde{\Phi}_{\text{flow}} \neq 1.00$ (Absorbing)

Forward/Backward Causal Taint
Wash Ring: $s_{\text{fwd}} \cdot s_{\text{bwd}} > 0.85$ (Bidirectional)
Benign: $s_{\text{fwd}} \cdot s_{\text{bwd}} \approx 0.02$ (Strictly Tree-like)

Hawkes Burst Timing Dispersion
Wash Ring: $\sigma_{\Delta t} < 2.4\,\text{h}$ (Coordinated Smurfing)
Benign: $\sigma_{\Delta t} > 48.0\,\text{h}$ (Stochastic Arrivals)

Transmission Capital Velocity
Wash Ring: $v_A = \text{mean}(A_e/\Delta t_e) \gg 10^3\,\$/\text{h}$
Benign: $v_A \approx 12.5\,\$/\text{h}$

THEORETICALLY SEPARABLE: Injective Multiset Mapping
12-D invariants strictly separate topologically isomorphic crime patterns!